$\sum_{n=0}^{+\infty} \left(\frac{2n+1}{n+3}\right)^n x^n \qquad a_n = \left(\frac{2n+1}{n+3}\right)^n$

$l = \lim_{n\to+\infty} \sqrt[n]{|a_n|} = l\left(|a_n|^{\frac{1}{n}}\right) = l\left[\left(\frac{2n+1}{n+3}\right)^n\right]^{\frac{1}{n}}$

$= l\left(\frac{2n+1}{n+3}\right) = 2$

thus $R = \frac{1}{l} = \frac{1}{2}$

thus the series converges on $\left]-\frac{1}{2}, \frac{1}{2}\right[$

Theorem the integer series $\sum_{n=0} a_n x^n$,

$\sum_{n=1}^{\infty} n a_n x^{n-1}$ and $\sum_{n=0} \frac{a_n}{n+1} x^{n+1}$ have the

same radius of convergence.

Theorem Let $\sum_{n=0}^{+\infty} a_n x^n$ be an integer

series with radius of convergence R and

it's sum $S(x) = \sum_{n=0}^{+\infty} a_n x^n$. Then

(i) the function S is continuous on $]-R, R[$

(ii) the function S admits antiderivatives

of the form $F(x) = \sum_{n=0}^{+\infty} \frac{a_n}{n+1} x^{n+1} + C \quad C \in \mathbb{R}$

$$= \sum_{n=0}^{+\infty} t^n = \sum_{n=0}^{+\infty} (-x)^n$$

$$\frac{1}{1+x} = \sum_{n=0}^{+\infty} (-1)^n x^n \quad , R=1$$

③ According to the previous example

$$\ln(1-x) = -\sum_{n=1}^{+\infty} \frac{x^n}{n} \quad , R=1$$

$$\int \frac{1}{1-x} = \int \sum_{n=0}^{+\infty} x^n$$

$$-\ln(1-x) = \sum_{n=0}^{+\infty} \frac{x^{n+1}}{n+1} \quad n=n+1 \quad = \sum_{n=1}^{+\infty} \frac{x^n}{n}$$

④ $\ln(1+x) = \ln(1-(-x)) = \ln(1-t) \quad t=-x$

$$= -\sum_{n=1}^{+\infty} \frac{t^n}{n} = -\sum_{n=1}^{+\infty} \frac{(-x)^n}{n} = -\sum_{n=1}^{+\infty} \frac{(-1)^n}{n} x^n$$

$$\ln(1+x) = \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{n} x^n \quad , R=1$$

⑤ $e^x = \sum_{n=1}^{+\infty} \frac{x^n}{n!} \quad , R=+\infty$

⑥ $\cos x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad , R=+\infty$

⑦ $\sin x = \sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad , R=+\infty$

Exercise Develop in integer series near 0 the function f

So $C=0$

We deduce that $S(x) = -\ln(1-x)$

i.e $\sum_{n=0}^{+\infty} \frac{x^n}{n} = -\ln(1-x)$, $x \in]-1,1[$

Integer series development

Definition to say that f is developpent into an integer series near 0 if there exists $\varepsilon > 0$ and a sequence $(a_n)_{n\geq 0}$ such as

$\forall x \in]-\varepsilon,\varepsilon[$, $f(x) = \sum_{n=0}^{+\infty} a_n x^n$

Definition The series $\sum_{n=0}^{+\infty} \frac{f^{(n)}(0)}{n!} x^n$ is called the taylor series of f near 0.

$f(x) = \sum_{n=0}^{+\infty} a_n x^n$ where $a_n = \frac{f^{(n)}(0)}{n!}$

Usual developpement

① $\frac{1}{1-x} = \sum_{n=0}^{+\infty} x^n$, $R=1$

② $\frac{1}{1+x} = \frac{1}{1-(-x)} = \frac{1}{1-t}$ $(t=-x)$

(iii) the function S is differentiable on $]-R,R[$ and $S'(x) = \sum_{n=1}^{+\infty} n a_n x^{n-1}$

Example Determine the sum $S(x)$ of the series $\sum_{n=1}^{\infty} \frac{x^n}{n}$ in $]-R,R[$.

Solution

$$l = \lim_{n\to+\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to+\infty} \left|\frac{\frac{1}{n+1}}{\frac{1}{n}}\right| = \lim_{n\to+\infty} \frac{n}{n+1} = 1$$

$R = \frac{1}{l} = \frac{1}{1} = 1$, so the series converges in $]-1,1[$ and it's sum $S(x) = \sum_{n=1}^{+\infty} \frac{x^n}{n}$

$\forall x \in]-1,1[$, $S'(x) = \sum_{n=1}^{+\infty} x^{n-1}$

$= x^0 + x^1 + x^2 + x^3 + \dots = \frac{1}{1-x} = S'(x)$

And consequently $S(x) = \int \frac{1}{1-x}\,dx$

$= -\ln(1-x) + C$

$S(0) = -\ln 1 + C \Rightarrow S(0) = C$

or $S(0) = \sum_{n=1}^{+\infty} \frac{0^n}{n} = \sum_{n=1}^{+\infty} 0 = 0$